

What is Mathematics?

A NAIVE DEFINITION, adequate for the dictionary and for an initial understanding, is that *mathematics is the science of quantity and space*. Expanding this definition a bit, one might add that mathematics also deals with the symbolism relating to quantity and to space.

This definition certainly has a historical basis and will serve us for a start, but it is one of the purposes of this work to modify and amplify it in a way that reflects the growth of the subject over the past several centuries and indicates the visions of various schools of mathematics as to what the subject ought to be.

The sciences of quantity and of space in their simpler forms are known as *arithmetic* and *geometry*. Arithmetic, as taught in grade school, is concerned with numbers of various sorts, and the rules for operations with numbers—addition, subtraction, and so forth. And it deals with situations in daily life where these operations are used.

Geometry is taught in the later grades. It is concerned in part with questions of spatial measurements. If I draw such a line and another such line, how far apart will their end points be? How many square inches are there in a rectangle 4 inches long and 8 inches wide? Geometry is also concerned with aspects of space that have a strong aesthetic appeal or a surprise element. For example, it tells us that in any parallelogram whatsoever, the diagonals bisect one another; in any triangle whatsoever, the three medians intersect in a common point. It teaches us that a floor can be

tilled with equilateral triangles or hexagons, but not with regular pentagons.

But geometry, if taught according to the arrangement laid out by Euclid in 300 B.C., has another vitally significant aspect. This is its presentation as a deductive science. Beginning with a number of elementary ideas which are assumed to be self-evident, and on the basis of a few definite rules of mathematical and logical manipulation, Euclidean geometry builds up a fabric of deductions of increasing complexity.

What is stressed in the teaching of elementary geometry is not only the spatial or visual aspect of the subject but the methodology wherein hypothesis leads to conclusion. This deductive process is known as *proof*. Euclidean geometry is the first example of a formalized deductive system and has become the model for all such systems. Geometry has been the great practice field for logical thinking, and the study of geometry has been held (rightly or wrongly) to provide the student with a basic training in such thinking.

Although the deductive aspects of arithmetic were clear to ancient mathematicians, these were not stressed either in teaching or in the creation of new mathematics until the 1800s. Indeed, as late as the 1950s one heard statements from secondary school teachers, reeling under the impact of the “new math,” to the effect that they had always thought geometry had “proof” while arithmetic and algebra did not.

With the increased emphasis placed on the deductive aspects of all branches of mathematics, C. S. Peirce in the middle of the nineteenth century, announced that “mathematics is the science of making necessary conclusions.” Conclusions about what? About quantity? About space? The content of mathematics is not defined by this definition; mathematics could be “about” anything as long as it is a subject that exhibits the pattern of assumption-deduction-conclusion. Sherlock Holmes remarks to Watson in *The Sign of Four* that “Detection is, or ought to be, an exact science and should be treated in the same cold and unemotional manner. You have attempted to tinge it with romanticism, which produces much the same effect as if you

worked a love-story or an elopement into the fifth proposition of Euclid.” Here Conan Doyle, with tongue in cheek, is asserting that criminal detection might very well be considered a branch of mathematics. Peirce would agree.

The definition of mathematics changes. Each generation and each thoughtful mathematician within a generation formulates a definition according to his lights. We shall examine a number of alternate formulations before we write *Finis* to this volume.

Further Readings. See Bibliography

A. Alexandroff; A. Kolmogoroff and M. Lawrentieff; R. Courant and H. Robbins; T. Danzig [1959]; H. Eves and C. Newsom; M. Gaffney and L. Steen; N. Goodman; E. Kasner and J. Newman; R. Kershner and L. Wilcox; M. Kline [1972]; A. Kolmogoroff; J. Newman [1956]; E. Snapper; E. Stabler; L. Steen [1978]

Where is Mathematics?

WHERE IS THE PLACE of mathematics? Where does it exist? On the printed page, of course, and prior to printing, on tablets or on papyri. Here is a mathematical book — take it in your hand; you have a palpable record of mathematics as an intellectual endeavor. But first it must exist in people’s minds, for a shelf of books doesn’t create mathematics. Mathematics exists on taped lectures, in computer memories and printed circuits. Should we say also that it resides in mathematical machines such as slide rules and cash registers and, as some believe, in the arrangement of the stones at Stonehenge? Should we say that it resides in the genes of the sunflower plant if that plant brings forth seeds arranged in Bernoullian spirals and transmits mathematical information from generation to generation? Should we say that mathematics exists on a wall if a lamp-

shade casts a parabolic shadow on that wall? Or do we believe that all these are mere shadow manifestations of the real mathematics which, as some philosophers have asserted, exists eternally and independently of this actualized universe, independently of all possible actualizations of a universe?

What is knowledge, mathematical or otherwise? In a correspondence with the writer, Sir Alfred Ayer suggests that one of the leading dreams of philosophy has been “to agree on a criterion for deciding what there is,” to which we might add, “and for deciding where it is to be found.”

The Mathematical Community

THERE IS HARDLY a culture, however primitive, which does not exhibit some rudimentary kind of mathematics. The mainstream of western mathematics as a systematic pursuit has its origin in Egypt and Mesopotamia. It spread to Greece and to the Graeco-Roman world. For some 500 years following the fall of Rome, the fire of mathematical creativeness was all but extinguished in Europe; it is thought to have been preserved in Persia. After some centuries of inactivity, the flame appeared again in the Islamic world and from there mathematical knowledge and enthusiasm spread through Sicily and Italy to the whole of Europe.

A rough timetable would be

Egyptian:	3000 B.C. to 1600 B.C.
Babylonian:	1700 B.C. to 300 B.C.
Greek:	600 B.C. to 200 B.C.
Graeco-Roman:	150 A.D. to 525 A.D.

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Islamic:	750 A.D. to 1450 A.D.
Western:	1100 A.D. to 1600 A.D.
Modern:	1600 A.D. to present.

Other streams of mathematical activity are the Chinese, the Japanese, the Hindu, and the Inca-Aztec. The interaction between western and eastern mathematics is a subject of scholarly investigation and conjecture.

At the present time, there is hardly a country in the world which is not creating new mathematics. Even the emerging nations, so called, all wish to establish up-to-date university programs in mathematics, and the hallmark of excellence is taken to be the research activity of their staffs.

In contrast to the relative isolation of early oriental and western mathematics from each other, the mathematics of today is unified. It is worked and transmitted in full and open knowledge. Personal secrecy like that practiced by the Renaissance and Baroque mathematicians hardly exists. There is a vast international network of publications; there are national and international open meetings and exchanges of scholars and students.

In all honesty, though, it should be admitted that restriction of information has occurred during wartime. There is also considerable literature on mathematical cryptography, as practiced by the professional cryptographers, which is not, for obvious reasons, generally available.

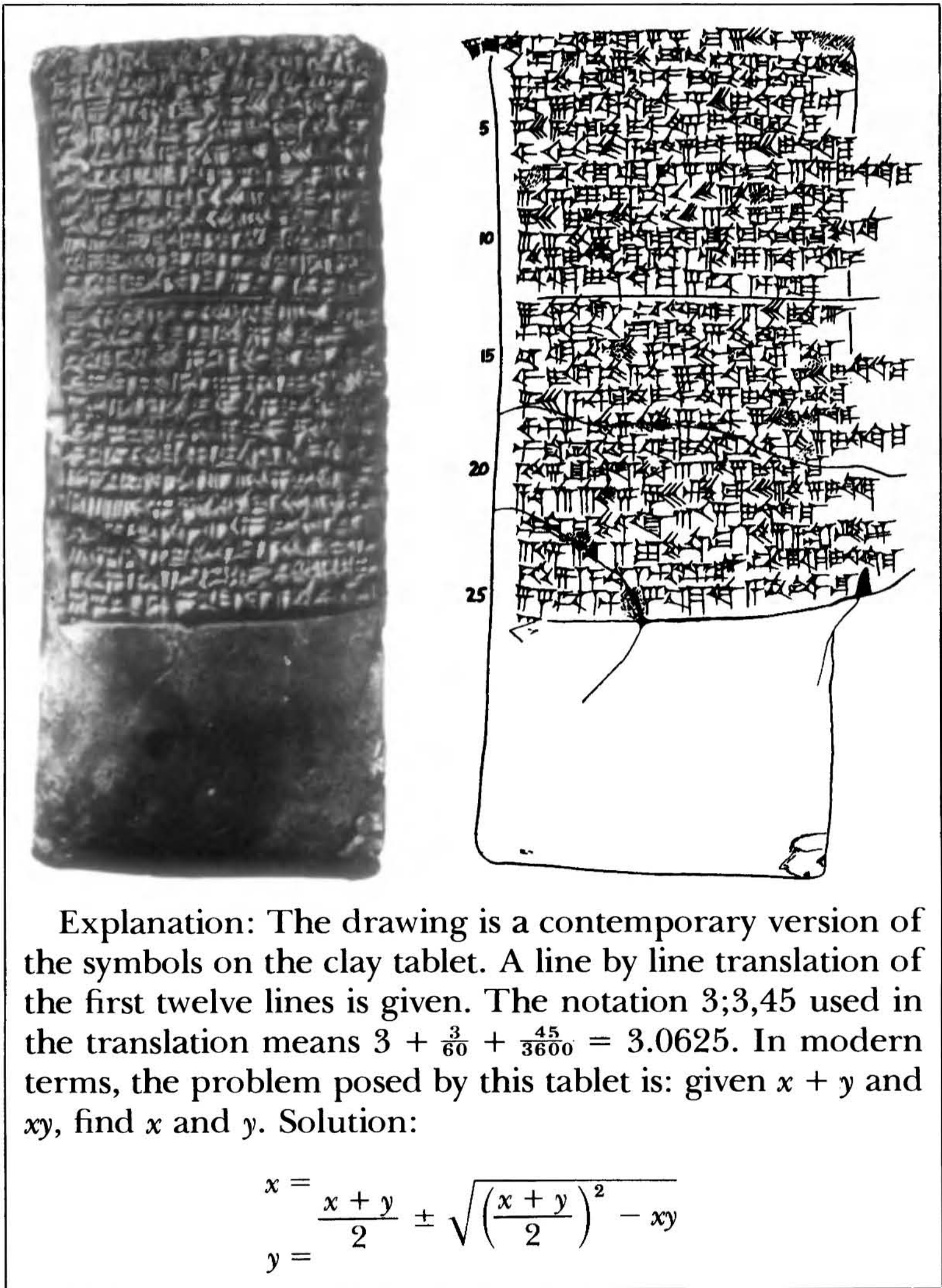
In the past mathematics has been pursued by people in various walks of life. Thomas Bradwardine (1325) was Archbishop of Canterbury. Ulugh Beg with his trigonometric tables was the grandson of Tamerlane. Luca Pacioli (1470) was a monk. Ferrari (1548) was a tax assessor. Cardano (1550) was professor of medicine. Viète (1580) was a lawyer in the royal privy council. Van Ceulen (1610) was a fencing master. Fermat (1635) was a lawyer. Many mathematicians earned part of their living as protégés of the Crown: John Dee, Kepler, Descartes, Euler; some even had the title of "Mathematicus." Up to about 1600, a mathematician could earn a few pounds by casting horoscopes or writing amulets for the wealthy.



François Viète
1540–1603



René Descartes
1596–1650



Explanation: The drawing is a contemporary version of the symbols on the clay tablet. A line by line translation of the first twelve lines is given. The notation 3;3,45 used in the translation means $3 + \frac{3}{60} + \frac{45}{3600} = 3.0625$. In modern terms, the problem posed by this tablet is: given $x + y$ and xy , find x and y . Solution:

$$x = \frac{x+y}{2} \pm \sqrt{\left(\frac{x+y}{2}\right)^2 - xy}$$

$$y = \frac{x+y}{2} \mp \sqrt{\left(\frac{x+y}{2}\right)^2 - xy}$$

These days there is nothing to prevent a wealthy person from pursuing mathematics full or part time in isolation, as in the era when science was an aristocrat's hobby. But this kind of activity is now not at a sufficiently high voltage to sustain invention of good quality. Nor does the church (or the monarchy) support mathematics as it once did.

For the past century, universities have been our princi-

What mathematics looked like in 1700 B.C.

Clay tablet with cuneiform writing from southern Iraq. The two problems that are worked out follow the standard procedure in Babylonian mathematics for quadratic equations.

¹ 9 (gín) is the (total expenses in) silver of a kilá; I added the length and the width, and (the result is) 6;30 (GAR); $\frac{1}{2}$ GAR is [its depth],

² 10 gín (volume) the assignment, 6 še (silver) the wages. What are the length (and) its width?

³ When you perform (the operations), take the reciprocal of the wages,

⁴ multiply by 9 gín, the (total expenses in) silver, (and) you will get 4,30;

⁵ multiply 4,30 by the assignment, (and) you will get 45;

⁶ take the reciprocal of its depth, multiply by 45, (and) you will get 7;30;

⁷ halve the length and the width which I added together, (and) you will get 3;15;

⁸ square 3;15, (and) you will get 10;33,45;

⁹ subtract 7;30 from 10;33,45, (and)

¹⁰ you will get 3;3,45; take its square root, (and)

¹¹ you will get 1;45; add it to the one, subtract it from the other, (and)

¹² you will get the length (and) the width. 5 (GAR) is the length; $1\frac{1}{2}$ GAR is the width.

Courtesy: Prof. A. J. Sachs, from Neugebauer and Sachs, "Mathematical Cuneiform Texts"

The Mathematical Landscape

pal sponsors. By releasing part of his time, the university encourages a lecturer to engage in mathematical research. At present, most mathematicians are supported directly or indirectly by the university, by corporations such as IBM, or by the federal government, which in 1977 spent about \$130,000,000 on mathematics of all sorts.

To the extent that all children learn some mathematics, and that a certain small fraction of mathematics is in the common language, the mathematics community and the community at large are identical. At the higher levels of practice, at the levels where new mathematics is created and transmitted, we are a fairly small community. The combined membership list of the American Mathematical Society, the Mathematical Association of America, and the Society for Industrial and Applied Mathematics for the year 1978 lists about 30,000 names. It is by no means necessary for one to think of oneself as a mathematician to operate at the highest mathematical levels; one might be a physicist, an engineer, a computer scientist, an economist, a geographer, a statistician or a psychologist. Perhaps the American mathematical community should be reckoned at 60 or 90 thousand with corresponding numbers in all the developed or developing countries.

Numerous regional, national, and international meetings are held periodically. There is lively activity in the writing and publishing of books at all levels, and there are more than 1600 individual technical journals to which it is appropriate to submit mathematical material.

These activities make up an international forum in which mathematics is perpetuated and innovated; in which discrepancies in practice and meaning are thrashed out.

Further Readings. See Bibliography

R. Archibald; E. Bell; B. Boos and M. Niss; C. Boyer; F. Cajori; J. S. Frame; R. Gillings; E. Husserl; M. Kline [1972]; U. Libbrecht; Y. Mikami; J. Needham; O. Neugebauer; O. Neugebauer and A. Sachs; D. Struik; B. Van der Waerden